

Problem

A TV signal is band-limited to 4.5 MHz. If samples are to be reconstructed at a distant point, what is the maximum sampling interval allowable?

Step-by-step solution

Step 1 of 1

We know that the nyquist rate of signal is

$$N_s \geq 2f_m$$

If sampled are spaced at nyquist rate, then they are recover without aliasing

$$N_s \geq 2 \times 4.5 \text{ MHz}$$

$$N_s \geq 9.0 \text{ MHz}$$

So, Maximum sampling interval allowable is 9.0MHz